When Variable Selection Meets Reinforcement Learning

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1. Introduction

Variable selection is a ubiquitous problem in predictive modeling. With increasing data collection capacity, many redundant and unnecessary features are often included in the datasets. It has been shown that careful variable selection approaches can eliminate noisy features and thus lead to model accuracy improvement. Moreover, having models with a small number of variables greatly helps to reduce model and data maintenance cost, and meanwhile enhances model interpretability and transparency.

However, the theoretically optimal approach, best subset selection, which enumerates all combinations of features, is infeasible even under a modest number of features p. Therefore, how to efficiently search the best subset of variables in this 2^p search space is not a trivial question. Common variable selection methods, such as forward/backward selection or Lasso, only visit a small subspace of the entire binary mask space, with strong assumptions about the optimal path, thus can be too restrictive.

To effectively search in this 2^p discrete space, we turn to Reinforcement Learning (RL) methods. In the literature, RL methods have been proposed for use in variable selection. Variable selection problems are modeled as Markov decision processes (MDP) and estimated with function-based RL method such as Monte Carlo method (Gaudel & Sebag, 2010), Temporal Difference (TD) (Fard et al., 2013) and DQN (Fang et al., 2019). Recently, they are also modeled as bandit and estimated with Thompson sampling (Liu & Ročková, 2023) and policy gradient method (Le et al., 2025).

In this paper, we will focus on policy gradient approaches, more specifically, Proximal Policy Optimization (PPO) with clipped objective, to conduct thorough analysis to compare these two options, bandit vs MDP, for variable selection. We will use multiple objective functions, including out-of-sample accuracy metrics, information criteria, and Bayes factor (see Section 2.2 for details) to learn with RL design works better for which objective functions. We will also compare the RL approaches with traditional model search methods such as forward/back selection, lasso, adaptive lasso, and MCMC (see Section 2.3 for details).

We will conduct simulation studies and real-world examples to conduct the above comparison. We notice that some existing papers' examples, both simulation and real data, are usually on the smaller side, e.g., sample sizes are often on the hundred, or at most thousand scale, which makes them seem like toy examples and make us wonder if the results generalize well to real use cases. So here, we plan to use real industry standard big data (Gorishniy et al., 2024). We will compare both variable selection accuracy (e.g., F1 score) on simulation data, and prediction accuracy (e.g., test RMSE) on real data.

2. Variable Selection in a Nutshell

2.1. Background and Notations

Let us consider normal (Gaussian) linear regression models. Suppose that among the p available predictive variables (features), denoted by $X_1, \ldots, X_p$, only a subset of them should be included in the regression model to predict a continuous target variable Y.

We can denote a binary vector $\boldsymbol{\gamma} \in \{0,1\}^p$ to indicate a subset model, which includes feature X_j if and only if $\gamma_j = 1$. Thus, on a training dataset with sample size n, the corresponding regression model can be written as

$$\mathbf{Y} = \beta_0 \mathbf{1}_n + \sum_{j:\gamma_j=1} \beta_j \mathbf{X}_j + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \overset{\text{iid}}{\sim} \mathrm{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n),$$

where β_0 is the intercept, β_j is the regression coefficient for variable X_j, and σ^2 is the normal variance of the residuals.

2.2. Common Objective Functions

For variable selection, suppose that the true data generating model $\boldsymbol{\gamma}^*$ is among the subset models being considered. Our goal is to find the true subset $\boldsymbol{\gamma}^*$ via optimization. Here, we consider several objective functions choices:

- **Cross-validated RMSE**: accesses out-of-sample performance to avoid the inclusion of redundant features

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or the risk of overfitting. We will minimize K-fold cross-validated RMSE.

- **Information criteria**: balances training accuracy and model complexity. We will minimize BIC and AIC:

$$\text{BIC}(\boldsymbol{\gamma}) = -\log \hat{L}_{\boldsymbol{\gamma}} + \frac{p_{\boldsymbol{\gamma}}}{2}\log(n),$$
$$\text{AIC}(\boldsymbol{\gamma}) = -\log \hat{L}_{\boldsymbol{\gamma}} + p_{\boldsymbol{\gamma}},$$

  where in model $\boldsymbol{\gamma}$, $\hat{L}_{\boldsymbol{\gamma}}$ is the maximized likelihood on the training data and $p_{\boldsymbol{\gamma}}$ is the number of features included.

- **Bayesian factor**: in the Bayesian paradigm, all parameters, including the subset model indicator $\boldsymbol{\gamma}$, can be viewed as random variables. Following the g-prior in Liang et al. (2008), we specify the prior distribution as follows.

$$p(\alpha, \sigma^2) \propto 1/\sigma^2$$
$$\boldsymbol{\beta}_{\boldsymbol{\gamma}} \mid \sigma^2, \boldsymbol{\gamma} \sim \text{N}\left(\mathbf{0}, g\,\sigma^2\ \left(\mathbf{X}_{\boldsymbol{\gamma}}^\top \mathbf{X}_{\boldsymbol{\gamma}}\right)^{-1}\right)$$

  Here, $g > 0$ is a hyper-parameter. A common choice is $g = \max\{p^2, n\}$. The advantage of g-prior is that we can integrate out the parameters $\alpha, \sigma^2, \boldsymbol{\beta}_{\boldsymbol{\gamma}}$ to obtain the marginal likelihood $p(\mathbf{Y} \mid \boldsymbol{\gamma}, g)$ in a closed form. Therefore, we can use the Bayes factor of model $\boldsymbol{\gamma}$ vs the null model (intercept only) $\boldsymbol{\gamma}_\emptyset$ as an objective function for $\boldsymbol{\gamma}$ and our goal is to maximize it.

$$\text{BF}_{\boldsymbol{\gamma}:\boldsymbol{\gamma}_\emptyset} = (1+g)^{\frac{n-p_{\boldsymbol{\gamma}}-1}{2}}\left[1 + g\left(1 - R^2_{\boldsymbol{\gamma}}\right)\right]^{-\frac{n-1}{2}},$$

  where $R^2_{\boldsymbol{\gamma}}$ is the coefficient of determination.

2.3. Common Search Methods

Here, we will not discuss feature screening as just using the marginal relationship between each feature and the target usually lead to suboptimal results or best subset selection due to its high computational cost.

- **Forward and/or backward selection**: In each iteration, add or remove a feature to optimize the objective function. These are greedy algorithms, which do not allow the case where an ok step may be followed by a very good step.

- **Lasso and adaptive lasso**: These are penalized likelihood methods that add an L1-norm term of the coefficients in the objective function to balance model accuracy and complexity. With the shrinkage parameter $\lambda > 0$ increases, Lasso shrinks more and more coefficients to exactly zero and thus achieves a shrinkage path with less and less variables. Adaptive Lasso enjoys this nice sparsity property while avoiding over-shrinking large signals that Lasso suffers from. The formula below is Lasso when $w_j = 1$ and Adaptive Lasso when $w_j = |\hat{\beta}_j|$, absolute values of the maximum likelihood estimators.

$$\arg\min_{\alpha,\boldsymbol{\beta}} \sum_i \left(y_i - \alpha - \mathbf{x}^\top \boldsymbol{\beta}\right)^2 + \lambda \sum_{j=1}^{p} \frac{|\beta_j|}{w_j}$$

- **Markov chain Monte Carlo**: a specific type of MCMC, Metropolis algorithm is common for Bayesian variable selection. In each iteration of this algorithm, it proposes a new state by flipping a random γ_j or swapping a random pair of $(\gamma_{j_1}, \gamma_{j_2})$. The new state will be accepted with probability, which equals the ratio of the posterior probabilities between the proposed state and the current state, capped at one.

3. Reinforcement Learning Methods for Variable Selection

In this paper, we will apply Proximal Policy Optimization (PPO) to select variables that optimize a certain objective function. The main benefit of taking the policy learning route is two-fold. First, no advanced search method needs to be explicitly designed (e.g. ϵ-greedy, tree search, etc). Second, stochastic policy fits the uncertain nature, especially for the Bayesian paradigm.

TODO: add minimal intro of PPO

3.1. The Bandit Way

Following the main idea in Le et al. (2025), we can model the variable selection as a bandit, i.e., a Markov decision process that terminates just after one time step. Thus, states and their value function are meaningless here, and our policy $\pi(a)$ does not depend on the state s.

An action $\mathbf{a}$ is a binary mask that represents a subset model,

$$\mathbf{a} = \boldsymbol{\gamma} \in \{0, 1\}^p,$$

where each dimension a_j for $j = 1, 2, \ldots, p$ follows a Bernoulli distribution. To estimate the policy distribution $\pi(\mathbf{a})$, our policy learning algorithm use a shallow fully-connected neural network, whose input is a constant vector and output is a p-dimensional vector of logits of the inclusion probabilities.

The reward will be the variable selection objective function discussed in Section 2.2. For the ones that we are supposed to minimize, such as CV-RMSE, AIC, and BIC, we use the negative of those metrics as our reward.

The bandit way is similar to the recent papers Le et al. (2025). The main difference is that we will use neural

network to model the policy instead of the independent beta distribution used in the paper. Because features are usually correlated, the paper's assumption that the inclusion of features are independent of each other is too restrictive.

3.2. The Markov Decision Process Way

To mimic the forward/backward selection idea, we can relax the termination after one step setting, by modeling variable selection via a Markov Decision Process (MDP) that terminates after N steps. Inspired by Gaudel & Sebag (2010), Fard et al. (2013), we let a state $\mathbf{s}$ be a binary mask that represents a subset model,

$$\mathbf{s} = \boldsymbol{\gamma} \in \{0, 1\}^p,$$

while an action a is to flip a feature (from 0 to 1 or from 1 to 0) with index a, plus a stop to terminate the process, i.e.,

$$a \in \mathcal{A} = \{1, 2, \ldots, p, \text{stop}\}.$$

Thus, in the $(\mathbf{s}, a, \mathbf{s}')$ tuple, if $1 \leq a \leq p$, then this is a forward selection step to include feature X_a if it is not inside $\mathbf{s}$, or a backward selection step to remove feature X_a if it is inside $\mathbf{s}$.

We use the same reward as in the bandit case. Since we only care about the return for a single model $\boldsymbol{\gamma}$ rather than the cumulative returns in the trajectory, we set the MDP's discount factor $\gamma = 0$ (We apologize for abusing the notation). Interestingly, Fard et al. (2013), in a similar MDP setting, choose this discount factor to be 0.3. We think this might help in identifying promising trajectories so we are open to this option, too.

4. Simulation Studies

4.1. Simulation Designs

5. Real Data Examples

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